

AE-616: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-610), kinetic

Date: 2025-11-11

Tags: Dihydroxo Radiation PNP P(tBu)N(py)P(tBu) [Ru(CO)(OH)2(PNP)] NMR AE 1H 31P Irradiation Power measurement photocatalysis

Category: Two-photon

Status: Done

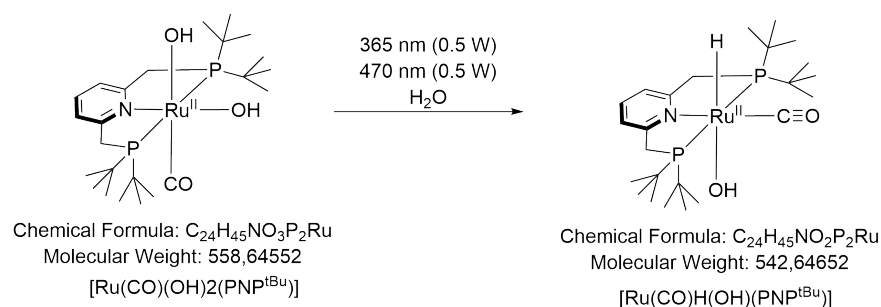
Created by: Alexander Eith

Objective

kinetic study of formation of hydride and side products during irradiation of [Ru(CO)(OH)2(PNP{tBu})]

Reaction scheme/sample structure

Radiation of Dihydroxo



ChemDraw File (linked): [AE-590.cdxml](#)

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Similar experiments	Two-photon - AE-543: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1), kinetic, max. conc Two-photon - AE-597: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1), kinetic, max. conc

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	Concentration [mM]	Obtained concentration [mM]
[Ru(CO)(OH)2(PNP{tBu})] (max. conc approx. 1.44 mM) in water	Two-photon - AE-610: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water	/	/	/	/	/	558.6455	/	0.60	max. conc (approx. 1.44 mM)	approx. 1.44

Irradiation Parameters

Power measured using [Power Meter - 843-R-USB + 919P-020-12](#) in [Equipment - Advanced power measurment setup V1.0](#)

	Name
Used Set-up	Equipment - Advanced irradiation setup V1.0 I
Irradiation set-up number	Equipment - Irradiation setup 2 (CEEC II, E004)

	Light Source Name	Used power source	Wavelength [nm]	Power Setting [mW]	Analogue Setting [-]
First light source	Light Source - UHP LED 365 nm-2	Power Sources - BLS-18000-1 1	365	500	9.42
Second light source	Light Source - UHP LED 470 nm-2	Power Sources - BLS-13000-1E	470	500	6.97

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration	1, 3, 20, 24 h

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using standard [Protocol - Schlenk Technique](#)

The synthesis/reactions were performed in the dark by either wrapping the reaction flasks in aluminum foil or turning the lab light and light in the fume hood off, when the flasks were uncovered.

The NMR tube was placed in the front holder of the holder for 2 NMR tubes

Date	Time	Step	Observations	Pictures
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06.11	14:55	A Young type NMR was prepared according to Protocol - Preparation of NMR Sample (from step 3) using 0.60 mL of AE-610 , a Prep work - AE-489: Capillary production containing D2O and H3PO4	AE-616	/
	15:20	The NMR sample an NMR was measured	AE-616-0H slightly yellow solution	AE-616-0H.jpg
10.11	09:05	The power of the LEDs were adjusted and the sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	before irradiation 1.jpg
	09:38	The irradiation was started	/	during irradiation 1.jpg
	10:38	The irradiation was stopped after 1h	slightly more orange	after irradiation 1.jpg
	09:20	An NMR was measured	AE-616-1H	AE-616-1H.jpg
11.11	09:44	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	before irradiation 2.jpg
	09:44	The irradiation was started	/	during irradiation 2.jpg
	12:44	The irradiation was stopped after 3h	more orange	after irradiation 2.jpg
	12:58	An NMR was measured	AE-616-4H	AE-616-4H.jpg
12.11	12:51	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	before irradiation 3.jpg
	12:51	The irradiation was started	/	during irradiation 3.jpg
13.11	08:51	The irradiation was stopped after 20h	no big change	after irradiation 3.jpg
	09:01	An NMR was measured	AE-616-24H	AE-618-24H.jpg
	17:17	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	/	/
	17:18	The irradiation was started	/	/

	17:21	The irradiation was stopped after 24h	/	/
	17:40	An NMR was measured	AE-616-48H , no big change, maybe slightly darker	AE-616-48H.jpg

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation
06.11	18:27	AE-616-0H	¹ H, ³¹ P{ ¹ H}quant NMR	IAAC 400 MHz	H2O	AE-616-0H.zip	/	AE-616-0H_10.nmrium	/	As expected, before irradiation, dihydroxo is main species
10.11	12:07	AE-616-1H	¹ H, ³¹ P{ ¹ H}quant NMR	IAAC 400 MHz	H2O	AE-616-1H.zip	/	AE-616-0H_10.nmrium	/	No big difference
11.11	13:41	AE-616-4H	¹ H, ³¹ P{ ¹ H}quant NMR	IAAC 400 MHz	H2O	AE-616-4H.zip	/	AE-616-0H_10.nmrium	/	Hydride observed
13.11	10:27	AE-616-24H	¹ H, ³¹ P{ ¹ H}quant NMR	IAAC 400 MHz	H2O	AE-616-24H.zip	/	AE-616-0H_10.nmrium	/	Significant conversion obtained
17.11	09:37	AE-616-48H	¹ H, ³¹ P{ ¹ H}quant NMR	IAAC 400 MHz	H2O	AE-616-48H.zip	/	AE-616-48H_10.nmrium	/	Full conversion
Sample		Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product, 57 ppm) [%]	Yield (soluble side products) [%]	Yield (other side products) [%]	Data		
AE-616-1H		0	0	96	0	4	0	AE-616_Yield_Determination.xlsx		
AE-616-4H		3	0	80	6	11	0	AE-616_Yield_Determination.xlsx		
AE-616-24H		21	0	9	46	11	14	AE-616_Yield_Determination.xlsx		
AE-616-48H		24	0	0	30	7	40	AE-616_Yield_Determination.xlsx		

Results

Worked, significant product after 4 h, full conversion after 48 h

Linked experiments

Prep work - [AE-489: Capillary production](#)

Prep work - [AE-549: Degassing of Milli-Q water](#)

Two-photon - [AE-543: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] \(AE-456-1\), kinetic, max. conc](#)

Two-photon - [AE-591: Stocksolution of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] in water](#)

Two-photon - [AE-597: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] \(AE-570-1\), kinetic, max. conc](#)

Two-photon - [AE-598: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] \(AE-570-1\), kinetic, 1/2 conc](#)

Two-photon - [AE-603: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] \(AE-570-1\), kinetic, max. conc, reproduction](#)

Two-photon - [AE-610: Stocksolution of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] in water](#)

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Equipment - [Advanced irradiation setup V1.0 I](#)

Equipment - [Advanced power measurment setup V1.0 I](#)

Equipment - [Manual irradiation setup](#)

Equipment - [Irradiation setup 2 \(CEEC II, E004\)](#)

Equipment - [Irradiation setup 4 \(CEEC II, E002\)](#)

Light Source - [UHP LED 470 nm-1](#)

Light Source - [UHP LED 365 nm-2](#)

Light Source - [UHP LED 470 nm-2](#)

Power Meter - [843-R-USB + 919P-020-12](#)

Power Sources - [BLS-13000-1E](#)

Power Sources - [BLS-18000-1 1](#)

Protocol - [Schlenk Technique](#)

Attached files

AE-616-48H.jpg

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AE-616_Yield_Determination.xlsx

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AE-616-48H_10.nmrium

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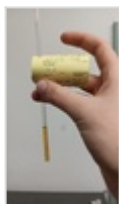
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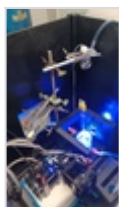
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after irrad 2.jpg

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during irrad 2.jpg

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before irradiation 2.jpg

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AE-616-1H.jpg

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after irradiation 1.jpg

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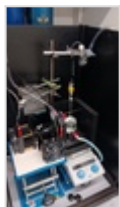
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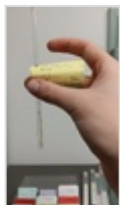
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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3410>